

# VAPOR PRESSURE OF THE METALLIC ELEMENTS

C. B. Alcock

This table gives coefficients in an equation for the vapor pressure of 65 metallic elements in both the solid and liquid state. Vapor pressures in the range  $10^{-10}$  to  $10^2$  Pa ( $10^{-15}$  to  $10^{-3}$  atm) are covered. The equation is:

$$\text{for } p \text{ in pascals: } \log(p/\text{Pa}) = 5.006 + A + BT^{-1} + C\log T + DT^{-3}$$

$$\text{for } p \text{ in atmospheres: } \log(p/\text{atm}) = A + BT^{-1} + C\log T + DT^{-3}, \text{ where } T \text{ is the temperature in K}$$

This equation reproduces the observed vapor pressures to an accuracy of  $\pm 5\%$  or better. Reprinted with permission of the publisher, Pergamon Press.

## REFERENCE

Alcock, C. B., Itkin, V. P., and Horrigan, M. K., *Canadian Metallurgical Quarterly*, 23, 309, 1984.

| Element,<br>state | A      | B      | C       | D       | Temperature<br>range |
|-------------------|--------|--------|---------|---------|----------------------|
| Li sol            | 5.667  | -8310  |         |         | 298-m.p.             |
| Li liq            | 5.055  | -8023  |         |         | m.p.-1000            |
| Na sol            | 5.298  | -5603  |         |         | 298-m.p.             |
| Na liq            | 4.704  | -5377  |         |         | m.p.-700             |
| K sol             | 4.961  | -4646  |         |         | 298-m.p.             |
| K liq             | 4.402  | -4453  |         |         | m.p.-600             |
| Rb sol            | 4.857  | -4215  |         |         | 298-m.p.             |
| Rb liq            | 4.312  | -4040  |         |         | m.p.-550             |
| Cs sol            | 4.711  | -3999  |         |         | 298-m.p.             |
| Cs liq            | 4.165  | -3830  |         |         | m.p.-550             |
| Be sol            | 8.042  | -17020 | -0.4440 |         | 298-m.p.             |
| Be liq            | 5.786  | -15731 |         |         | m.p.-1800            |
| Mg sol            | 8.489  | -7813  | -0.8253 |         | 298-m.p.             |
| Ca sol            | 10.127 | -9517  | -1.4030 |         | 298-m.p.             |
| Sr sol            | 9.226  | -8572  | -1.1926 |         | 298-m.p.             |
| Ba sol            | 12.405 | -9690  | -2.2890 |         | 298-m.p.             |
| Ba liq            | 4.007  | -8163  |         |         | m.p.-1200            |
| Al sol            | 9.459  | -17342 | -0.7927 |         | 298-m.p.             |
| Al liq            | 5.911  | -16211 |         |         | m.p.-1800            |
| Ga sol            | 6.657  | -14208 |         |         | 298-m.p.             |
| Ga liq            | 6.754  | -13984 | -0.3413 |         | m.p.-1600            |
| In sol            | 5.991  | -12548 |         |         | 298-m.p.             |
| In liq            | 5.374  | -12276 |         |         | m.p.-1500            |
| Tl sol            | 5.971  | -9447  |         |         | 298-m.p.             |
| Tl liq            | 5.259  | -9037  |         |         | m.p.-1100            |
| Sn sol            | 6.036  | -15710 |         |         | 298-m.p.             |
| Sn liq            | 5.262  | -15332 |         |         | m.p.-1850            |
| Pb sol            | 5.643  | -10143 |         |         | 298-m.p.             |
| Pb liq            | 4.911  | -9701  |         |         | m.p.-1200            |
| Sc sol            | 6.650  | -19721 | 0.2885  | -0.3663 | 298-m.p.             |
| Sc liq            | 5.795  | -17681 |         |         | m.p.-2000            |
| Y sol             | 9.735  | -22306 | -0.8705 |         | 298-m.p.             |
| Y liq             | 5.795  | -20341 |         |         | m.p.-2300            |
| La sol            | 7.463  | -22551 | -0.3142 |         | 298-m.p.             |
| La liq            | 5.911  | -21855 |         |         | m.p.-2450            |
| Ti sol            | 11.925 | -24991 | -1.3376 |         | 298-m.p.             |
| Ti liq            | 6.358  | -22747 |         |         | m.p.-2400            |
| Zr sol            | 10.008 | -31512 | -0.7890 |         | 298-m.p.             |
| Zr liq            | 6.806  | -30295 |         |         | m.p.-2500            |
| Hf sol            | 9.445  | -32482 | -0.6735 |         | 298-m.p.             |
| V sol             | 9.744  | -27132 | -0.5501 |         | 298-m.p.             |

# VAPOR PRESSURE OF THE METALLIC ELEMENTS (continued)

| Element,<br>state | A       | B      | C        | D       | Temperature<br>range |
|-------------------|---------|--------|----------|---------|----------------------|
| V liq             | 6.929   | -25011 |          |         | m.p.-2500            |
| Nb sol            | 8.822   | -37818 | -0.2575  |         | 298-2500             |
| Ta sol            | 16.807  | -41346 | -3.2152  | 0.7437  | 248-2500             |
| Cr sol            | 6.800   | -20733 | 0.4391   | -0.4094 | 298-2000             |
| Mo sol            | 11.529  | -34626 | -1.1331  |         | 298-2500             |
| W sol             | 2.945   | -44094 | 1.3677   |         | 298-2350             |
| W sol             | -54.527 | -57687 | -12.2231 |         | 2200-2500            |
| Mn sol            | 12.805  | -15097 | -1.7896  |         | 298-m.p.             |
| Re sol            | 11.543  | -40726 | -1.1629  |         | 298-2500             |
| Fe sol            | 7.100   | -21723 | 0.4536   | -0.5846 | 298-m.p.             |
| Fe liq            | 6.347   | -19574 |          |         | m.p.-2100            |
| Ru sol            | 9.755   | -34154 | -0.4723  |         | 298-m.p.             |
| Os sol            | 9.419   | -41198 | -0.3896  |         | 298-2500             |
| Co sol            | 10.976  | -22576 | -1.0280  |         | 298-m.p.             |
| Co liq            | 6.488   | -20578 |          |         | m.p.-2150            |
| Rh sol            | 10.168  | -29010 | -0.7068  |         | 298-m.p.             |
| Rh liq            | 6.802   | -26792 |          |         | m.p.-2500            |
| Ir sol            | 10.506  | -35099 | -0.7500  |         | 298-2500             |
| Ni sol            | 10.557  | -22606 | -0.8717  |         | 298-m.p.             |
| Ni liq            | 6.666   | -20765 |          |         | m.p.-2150            |
| Pd sol            | 9.502   | -19813 | -0.9258  |         | 298-m.p.             |
| Pd liq            | 5.426   | -17899 |          |         | m.p.-2100            |
| Pt sol            | 4.882   | -29387 | 1.1039   | -0.4527 | 298-m.p.             |
| Pt liq            | 6.386   | -26856 |          |         | m.p.-2500            |
| Cu sol            | 9.123   | -17748 | -0.7317  |         | 298-m.p.             |
| Cu liq            | 5.849   | -16415 |          |         | m.p.-1850            |
| Ag sol            | 9.127   | -14999 | -0.7845  |         | 298-m.p.             |
| Ag liq            | 5.752   | -13827 |          |         | m.p.-1600            |
| Au sol            | 9.152   | -19343 | -0.7479  |         | 298-m.p.             |
| Au liq            | 5.832   | -18024 |          |         | m.p.-2050            |
| Zn sol            | 6.102   | -6776  |          |         | 298-m.p.             |
| Zn liq            | 5.378   | -6286  |          |         | m.p.-750             |
| Cd sol            | 5.939   | -5799  |          |         | 298-m.p.             |
| Cd liq            | 5.242   | -5392  |          |         | m.p.-650             |
| Hg liq            | 5.116   | -3190  |          |         | 298-400              |
| Ce sol            | 6.139   | -21752 |          |         | 298-m.p.             |
| Ce liq            | 5.611   | -21200 |          |         | m.p.-2450            |
| Pr sol            | 8.859   | -18720 | -0.9512  |         | 298-m.p.             |
| Pr liq            | 4.772   | -17315 |          |         | m.p.-2200            |
| Nd sol            | 8.996   | -17264 | -0.9519  |         | 298-m.p.             |
| Nd liq            | 4.912   | -15824 |          |         | m.p.-2000            |
| Sm sol            | 9.988   | -11034 | -1.3287  |         | 298-m.p.             |
| Eu sol            | 9.240   | -9459  | -1.1661  |         | 298-m.p.             |
| Gd sol            | 8.344   | -20861 | -0.5775  |         | 298-m.p.             |
| Gd liq            | 5.557   | -19389 |          |         | m.p.-2250            |
| Tb sol            | 9.510   | -20457 | -0.9247  |         | 298-m.p.             |
| Tb liq            | 5.411   | -18639 |          |         | m.p.-2200            |
| Dy sol            | 9.579   | -15336 | -1.1114  |         | 298-m.p.             |
| Ho sol            | 9.785   | -15899 | -1.1753  |         | 298-m.p.             |
| Er sol            | 9.916   | -16642 | -1.2154  |         | 298-m.p.             |
| Er liq            | 4.668   | -14380 |          |         | m.p.-1900            |
| Tm sol            | 8.882   | -12270 | -0.9564  |         | 298-1400             |
| Yb sol            | 9.111   | -8111  | -1.0849  |         | 298-900              |
| Lu sol            | 8.793   | -22423 | -0.6200  |         | 298-m.p.             |
| Lu liq            | 5.648   | -20302 |          |         | m.p.-2350            |
| Th sol            | 8.668   | -31483 | -0.5288  |         | 298-m.p.             |
| Th liq            | -18.453 | -24569 | 6.6473   |         | m.p.-2500            |
| Pa sol            | 10.552  | -34869 | -1.0075  |         | 298-m.p.             |

# VAPOR PRESSURE OF THE METALLIC ELEMENTS (continued)

| Element,<br>state | A      | B      | C       | D       | Temperature<br>range |
|-------------------|--------|--------|---------|---------|----------------------|
| Pa liq            | 6.177  | -32874 |         |         | m.p.-2500            |
| U sol             | 0.770  | -27729 | 2.6982  | -1.5471 | 298-m.p.             |
| U liq             | 20.735 | -28776 | -4.0962 |         | m.p.-2500            |
| Np sol            | 19.643 | -24886 | -3.9991 |         | 298-m.p.             |
| Np liq            | 10.076 | -23378 | -1.3250 |         | m.p.-2500            |
| Pu sol            | 26.160 | -19162 | -6.6675 |         | 298-600              |
| Pu sol            | 18.858 | -18460 | -4.4720 |         | 500-m.p.             |
| Pu liq            | 3.666  | -16658 |         |         | m.p.-2450            |
| Am sol            | 11.311 | -15059 | -1.3449 |         | 298-m.p.             |
| Cm sol            | 8.369  | -20364 | -0.5770 |         | 298-m.p.             |
| Cm liq            | 5.223  | -18292 |         |         | m.p.-2200            |